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CS 499 Milestone Three: Enhancement Two — Algorithms and Data Structures

Artifact Description

introduces my third project for CS 360, an inventory application developed for the course Mobile Architecture and Programming. Built for the Android platform using Java and XML, the application supports user login, inventory management, local storage, and SMS notifications.

Justification for Inclusion in the ePortfolio

management application in my electronic portfolio, as it verifies that I am capable of optimizing existing applications using algorithms and data structures. While the original application supported create, read, update, and delete (CRUD) operations and data display, it could not adapt to the demand for efficient retrieval following the system's capacity expansion.

I independently iterated on this small Java inventory management application. At the underlying implementation level, I created a new InventoryItem class with custom attributes, updated the InventoryActivity to use ArrayList for inventory management, and added five new functions including search and sorting. This work not only improved the application's usability, but also proved that I have mastered core development capabilities such as list operations.

Course Outcomes

This enhancement supports Course Outcome 3 because I improved the application using algorithmic principles and considered how users can access inventory data more efficiently. The search and low-stock filter operations use a linear scan of the inventory list, which is appropriate for a small mobile inventory application. This enhancement also supports Course Outcome 4 because I used Java, Android Studio, object-oriented programming, and data structures to deliver useful new functionality.

My outcome-coverage plan has not changed. I used my first enhancement to demonstrate software design and engineering, and this second enhancement demonstrates algorithms and data structures. My next enhancement will focus on improving the database features of the same application.

Reflection

While completing this enhancement, I learned that an application becomes more useful when users can organize and locate data quickly. The original app stored inventory information, but it did not help users search or compare records easily. Adding search, sorting, and low-stock filtering improved both the functionality and the user experience.

When developing a personal efficiency application for my own use, I encountered a core transformation challenge: I needed to replace the scattered plain-text records in the original application with InventoryItem objects for unified processing. This requirement forced me to sort out the logic of the entire data workflow, and also deepened my understanding of the role data structures play in supporting professional software development.

This enhancement also demonstrates the use of Java collections because ArrayList stores inventory records in sequence, and Comparator provides the rules needed to sort custom InventoryItem objects by name or quantity (Oracle, n.d.-a; Oracle, n.d.-b).

References

Oracle. (n.d.-a). ArrayList (Java Platform SE 8). Java documentation. https://docs.oracle.com/javase/8/docs/api/java/util/ArrayList.html

Oracle. (n.d.-b). Comparator (Java Platform SE 8). Java documentation. https://docs.oracle.com/javase/8/docs/api/java/util/Comparator.html